

The Seven Post-Meeting Tasks

2026-07-29

1 The Moral Anchor: The Two Assumptions Compared

Meeting Note: The new moral anchor assumption is more complex. What do we really gain from it? We should adopt the new assumption only if the gain is real.

(i) The two assumptions. The moral term is $-\frac{\mu}{2}(t - t^e)^2$ in every game; the two versions of the model differ only in what the anchor t^e is in the trust game. *Old*: the equal-split action, $t^e = \frac{1}{2}$ —a norm about the *action*. *New*: the send that equalizes expected payoffs at the believed returned share s , $t^e(s) = [1 + (1 + r)(1 - 2s)]^{-1}$ —a norm about *payoffs*, evaluated using the sender’s beliefs. In the DG and UG (where $r = 0$) the two coincide, so the entire comparison lives in the trust game: $t = \frac{1}{2}$ equalizes payoffs there only if the receiver returns exactly a third of the proceeds.

(ii) Comparative statics: do the directional/qualitative predictions differ?

Comparative static of the TG send	Expression	Old anchor	New anchor
$\partial t / \partial (\rho / \mu)$	r	+	+
$\partial t / \partial (\sigma / \mu)$	$-(1 + r)(1 - s)$	−	−
$\partial t / \partial s$ for a category with $\sigma / \mu > 0$	$dt^e / ds + (1 + r) \sigma / \mu$	+	+
$\partial t / \partial s$ for a category with $\sigma / \mu = 0$	dt^e / ds	0	+

$\partial t / \partial s$ is the TG sender’s belief sensitivity: the response of her action to her believed returned share (the share of the tripled proceeds she expects back). In the DG and UG the two anchors coincide, so no prediction of any kind differs there. In the trust game, both anchors predict that more attention to the surplus raises the send, that more attention to own earnings lowers it, and—for any category with $\sigma / \mu > 0$ —that a higher believed return raises it, since beliefs enter the material term either way. The signs split only at $\sigma / \mu = 0$: an old-anchor sender with no attention to own earnings has no reason to condition on beliefs at all, while under the new anchor the norm itself moves with the belief, so *every* sender is belief-sensitive. (A second qualitative difference, the full-send corner, appears in (iv) below.)

(iii) Evidence on the differing comparative statics prediction. There is a category where the calibrated σ / μ is 0 and is Mutual Benefit/Cooperation (MBC). For MBC, the old anchor predicts a belief sensitivity of zero, while the new anchor predicts a positive belief sensitivity. The OLS slope of the sender’s action on the sender’s hypothetical believed return share in the control sample (a model-free estimate) is positive, +0.264 (se 0.098, $N = 170$). The one comparative-statics difference between the anchors favors the new one.

(iv) Other evidence. *Qualitative, no calibrated parameters.* First, senders who invest everything. Under the new anchor, $t^e(s) \geq 1$ for $s \geq \frac{1}{2}$, that is, a sender who expects at least half of the proceeds back has

a norm anchor at or beyond the full send, so full sends should concentrate among such optimists—while the old anchor holds norm-driven senders at one half regardless of beliefs and cannot account for moral optimists sending everything. In control, 30.5% of trust-game senders hold beliefs $s \geq \frac{1}{2}$, 52.0% under the market frame—and that is where the spike at the full send appears in the distribution of sends. Second, the receiver side. Realized returned shares among Moral-good and MBC receivers rise with the amount sent, tracking the equal-payoff schedule.

Levels, computed at the calibrated parameters. Here we compare predicted versus estimated magnitudes of the belief sensitivity of the TG send. The predicted columns evaluate the $\partial t / \partial s$ rows of the table in (ii) using the calibrated σ/μ and the measured believed share s . The estimated column is the OLS slope as in (iii).

Category	Predicted, old anchor	Predicted, new anchor	Estimated	Estimated/ pred. old	Estimated/ pred. new
Moral	0.113	1.461	0.306	2.71	0.21
Self-interest	1.192	2.029	0.451	0.38	0.22
Mutual Benefit/Coop. (MBC)	0.000	1.742	0.264	∞	0.15

Under the old anchor, beliefs matter only through the material term; moral senders are estimated to be 2.7 times *more* belief-sensitive than predicted; and the three estimated-to-predicted ratios are wildly different from one another (2.71 / 0.38 / ∞). Under the new anchor, the norm channel dr^e/ds enters every category's sensitivity, and the three ratios become uniform (0.15–0.22)—the attenuation one expects when coarse categories measure the underlying weights with error.

2 Calibration Variants: Medians, Alternative Anchor, Which Beliefs, MBC

Meeting Notes: prepare alternative calibrations using (a) median rather than mean choices; (b) the equal-split action anchor; (c) the other measured belief or both measured beliefs.

The standing calibration. For reference, the paper's calibration table (Table 4) under the standing methodology—category means as moments, hypothetical beliefs, the equal-payoff anchor:

Category	Attention		Beliefs			Acceptance at chosen offer	
	σ/μ	ρ/μ	a	b	s	implied	measured
Moral	0.038 (0.007)	0.022 (0.013)	–	–	0.315	0.565	0.795
Self-interest	0.397 (0.012)	0.426 (0.027)	0.324 (0.042)	0.481 (0.120)	0.220	0.525	0.701
Mutual Benefit/Coop.	0.000 (0.044)	0.048 (0.047)	0.324 [†]	0.481 [†]	0.357	0.598	0.711

Notes: σ/μ , ρ/μ : attention to own earnings and to the joint surplus, relative to the moral weight; (a, b) : believed acceptance schedule $a + bt$, unidentified for Moral and [†]imposed on MBC from Self-interest; s : believed returned share at the reference send (measured, not calibrated); the last two columns are the overidentifying test at chosen offers. Bootstrap standard errors in parentheses: participants resampled within game ($B = 1,000$), the full calibration rerun on each draw; absent where the parameter is unidentified (Moral's schedule) or imposed (MBC's).

(a) Median choices instead of mean choices. The action distributions are bimodal, so their medians sit on the modes, and the calibration inputs land on corner values at which inverting an interior first-order condition is uninformative:

Category	t_{DG}		t_{UG}		t_{TG}	
	mean	median	mean	median	mean	median
Moral	0.462	0.500	0.500	0.500	0.441	0.417
Self-interest	0.103	0.000	0.417	0.458	0.295	0.250
Mutual Benefit/Coop.	0.333	0.500	0.568	0.500	0.635	0.500

Consequences: Moral’s σ/μ becomes exactly 0 (the median Moral dictator chooses the equal split) and Self-interest’s exactly 0.5, the maximum (the median Self-interest dictator gives nothing); with the median Moral offer exactly at one half, the schedule inversion returns $b = 0$; no median variant yields a valid acceptance schedule in any category ($b \leq 0$ or $a + b > 1$ throughout); and Moral’s ρ/μ turns slightly negative (-0.03).

(b) First-order conditions from the equal-split action anchor. Nothing of substance changes except the TG belief sensitivities, which is exactly the comparison of Section 1.

(c) Two beliefs: which to use. Each first mover reports two beliefs (one at her chosen action, one at the common one-third reference action). Three ways of feeding these into the UG calibration were computed:

Route	Schedule (Self-int.)	Valid?	Hold-out check
Reference belief + FOC (current)	$a = 0.32, b = 0.48$	yes ($a+b = 0.81$)	misses chosen-action beliefs by +11 to +23pp (= the optimism finding)
Chosen-action belief + FOC	$a = 0.39, b = 0.75$	no ($a+b = 1.14$; Moral $a = -3.8$)	misses the reference belief by -15pp (mirror image)
Both belief points, no FOC	$a = -0.38, b = 2.60$	no ($a < 0$; $a+b$ up to 2.6)	—

Only the reference-belief route yields a valid probability schedule. Chosen-action beliefs are unusable as calibration inputs because they are systematically too optimistic—and that is precisely what makes them valuable as the held-out check, whose failure becomes the forecast-error result of §4.3. On the trust-game side the choice is immaterial: replacing the reference believed share with the chosen-action one moves Self-interest’s ρ/μ from 0.426 to 0.427 and changes nothing qualitative.

3 Structural Similarity with a More Neutral Prompt

Meeting Notes: re-compute similarity between stories and games (previously Table 4, now Table 5) with a more neutral prompt (that is, not instructing LLM raters to focus on “focus on the structural and strategic features of the situation, not on surface thematic content”)

What this exercise measures. Three frontier models (GPT-5.5 Thinking, Claude Opus 4.7, Gemini 3 Pro) rated, on a 0–100 scale, the similarity between each of the two *stories* (Bonus, Aid) and each of the four control *games instructions* (DG-KW, DG-LT, UG, TG). Each model rated the eight story–game pairs under neutral labels (Games A–D, Story 1/Story 2), with three label permutations per model. After the ratings (Prompt 1), the same conversations produced the five-dimension structural decomposition (Prompt 2) and the retrieval split (Prompt 3: which story a participant who has just read the game instructions would more likely call to mind, a 100-point split with no restriction on what may drive recall).

The concern. The May prompt instructed the raters to “focus on the structural and strategic features of the situation, not on surface thematic content” and listed five features to consider. This raises two worries: the instruction tells the rater to ignore precisely the dimensions on which Bonus and Aid differ, so the

Bonus $\approx$ Aid result could be built into the prompt; and the feature list imposes our own taxonomy, so the similarity ordering DG-KW > DG-LT > UG > TG could reflect the instructions rather than the situations.

The revised prompt. In place of the guidance, the prompt reads: “Use your own judgment to decide which aspects of the situations are relevant and how much weight to give them.”

Models. Opus 4.7 for Claude and ChatGPT 5.5 Thinking for ChatGPT, *the same models as in May*, so those two comparisons isolate the prompt change; Gemini 3 Pro is no longer offered, so we used Gemini Pro 3.1—the one arm whose comparison bundles the prompt change with a model update.

Results. The May findings survive the neutral prompt.

Game	Structural similarity (0–100)			Retrieval split
	Bonus	Aid	Mean	Bonus / Aid
<i>Panel A. May run, guided prompt, three models $\times$ three permutations (= paper Table 5)</i>				
DG-KW	94.6	94.7	94.6	41 / 59
DG-LT	51.8	50.7	51.2	54 / 46
UG	30.1	29.8	29.9	60 / 40
TG	15.0	13.9	14.4	72 / 28
<i>Panel B. July rerun, neutral prompt, three arms $\times$ three permutations</i>				
DG-KW	90.8	90.3	90.6	46 / 54
DG-LT	46.6	46.2	46.4	48 / 52
UG	32.4	29.3	30.9	62 / 38
TG	15.6	14.2	14.9	67 / 33

4 Similarity Between Games and Receivers’ Vignettes

Meeting Notes: Add a table with similarity between games and receivers’ vignettes.

Why this required new data. We generated 24 sender-based vignettes and 8 receiver-based vignettes but presented results only for similarity ratings for the receiver vignettes.

Results. Under the market frame the pictured ultimatum-game counterpart moves toward the accepting type and away from the rejecting type—the same direction as first movers’ measured acceptance beliefs—while the trust game barely moves, consistent with the similarity ratings for sender vignettes.

UG Contexts	Accepting	Rejecting	TG Contexts	Returning	Keeping
Control	47.0	58.8	Control	62.5	42.5
Market	50.0	56.7	Market	61.0	43.5
Market – Control	+3.0	–2.2		–1.5	+1.0

5 A Richer and More Diverse Vignette Pool for the Similarity Exercise

Meeting Notes: for similarity, we need some vignettes whose characters are (moral, selfish, cooperative) firms rather than people; in general, we need a more diverse set of vignettes.

The new set of 48 vignettes. One cell for every combination of attention category (Moral, Self-interest, Mutual Benefit/Cooperation) and sender belief (high or low: the protagonist expects the counterpart to respond favorably, or not), six cells in all, with eight vignettes per cell: $3 \times 2 \times 8 = 48$. Within every cell, four vignettes describe interactions among individuals and four involve firms or entrepreneurs, so neither the category nor the belief level is confounded with the character type.

Methodology. Rater: Opus 4.8. Three fresh conversations, each with independently permuted neutral labels. In each conversation the rater first reads the 48 vignettes; then, for each of the ten context texts (the two stories and the Control and Market instructions of the four games), rates the context’s similarity to every sender vignette on a 0–100 scale and distributes 100 retrieval points across the vignettes.

Results. The category similarities are the mean ratings over a category’s sixteen vignettes (its eight high-belief and eight low-belief), averaged over the three conversations. The belief margins are the difference between the mean rating of a category’s eight high-belief vignettes and the mean rating of its eight low-belief vignettes, that is, a similarity measure for the belief component of the representation. Each cell reports the Market value minus the Control value of these statistics, by game.

	Control			Market		
	Moral	Self-Interest	MBC	Moral	Self-Interest	MBC
<i>Category Similarity</i>						
DG-KW	45.4	12.0	10.8	19.8	28.7	15.7
DG-LT	40.7	12.6	11.8	19.8	28.6	15.7
UG	35.2	13.6	14.6	15.6	30.9	18.5
TG	15.4	28.9	42.8	14.3	30.9	41.4
Bonus	42.1	7.1	7.2			
Aid	35.7	7.9	7.4			
<i>Belief Margin (High – Low)</i>						
DG-KW	–3.5	+0.2	+0.1	–0.7	+9.3	–0.0
DG-LT	–3.4	+0.1	+0.4	–0.7	+9.5	–0.0
UG	–2.5	+0.3	+0.1	+0.5	+8.5	–0.4
TG	+1.8	–3.0	–0.9	+1.3	–2.8	–1.3
Bonus	+0.2	+0.3	+0.5			
Aid	–2.8	–1.0	–0.1			

Four findings. First, the new set reproduces the structure we obtained with the original sets: Moral is the most similar category for the control fixed-pie games, Mutual Benefit dominates the control trust game, and the market frame crowds out Moral in the fixed-pie games while raising Self-interest. Second, the belief margin detects, in similarity space alone, the optimism the market frame induces. Recall what the margin measures: a positive Self-interest margin says the context resembles the situations in which a self-interested protagonist *expects the counterpart to go along* (accept the terms, repay the advance) more than the situations in which she expects the opposite. In the control fixed-pie games this margin is essentially zero (+0.1 to +0.3): the abstract descriptions are no more similar to confident self-interest than to wary self-interest. Under the market frame it jumps to +8.5–+9.5: the market versions read specifically like *confident* self-interest. That is the direction of the measured beliefs—market proposers hold higher acceptance beliefs—so the similarity module now sees the belief component of the representation move with the frame, without any belief being elicited. Third, the trust game stays flat on *both* margins: even the belief-augmented similarity measure detects no market shift there, so the trust-game anomaly (representations and beliefs move; similarity does not) is robust to the richer set, which strengthens the interpretation that the market frame works on the trust game through the availability of market experience rather than through resemblance of

the described situation. Fourth: the Moral belief margin is *negative* in the control fixed-pie contexts (-2.5 to -3.5)—read literally, the dictator and ultimatum descriptions sit closer to the *low*-belief Moral vignettes than to the high-belief ones. The reason is mechanical. Giving the protagonist a belief requires a counterpart who can respond, so the Moral high-belief vignettes feature counterparts who repay, honor agreements, reciprocate—and thereby read like trust-game situations; a one-sided context, whose recipient can do nothing, matches the low-belief versions better whatever its moral content.

6 Decomposition in Table 14 with Actual-Choice Beliefs

Meeting Note: Decomposition using actual-choice (rather than hypothetical) beliefs.

What we did. A second specification of the preregistered decomposition in Table 14, with representation cells defined by the stated category interacted with terciles of the *actual* (chosen-action) belief, next to the original specification with hypothetical beliefs.

Comparison	Game	Δ mean	Hypothetical beliefs			Actual beliefs			N
			Repr.	Behav.	Repr. %	Repr.	Behav.	Repr. %	
Market vs Control	DG-KW	−0.101 (0.018)	−0.039 (0.016)	−0.061 (0.020)	39%	−0.039 (0.016)	−0.061 (0.020)	39%	728
Market vs Control	UG	−0.220 (0.013)	−0.036 (0.015)	−0.184 (0.020)	16%	−0.024 (0.012)	−0.196 (0.018)	11%	1027
Market vs Control	TG	+0.133 (0.019)	+0.078 (0.012)	+0.054 (0.018)	59%	+0.085 (0.013)	+0.048 (0.017)	64%	970
Aid vs Bonus	DG-KW	−0.059 (0.013)	−0.039 (0.010)	−0.021 (0.008)	65%	−0.039 (0.010)	−0.021 (0.008)	65%	790
Aid vs Bonus	UG	−0.024 (0.007)	−0.012 (0.003)	−0.011 (0.006)	52%	−0.016 (0.004)	−0.008 (0.006)	68%	1120
Aid vs Bonus	TG	+0.008 (0.020)	+0.011 (0.014)	−0.003 (0.016)	—	−0.003 (0.015)	+0.011 (0.014)	—	937

Note: symmetrized (Shapley/Oaxaca–Blinder) decomposition of the treatment effect on mean Player-1 actions into a representation component and a behavior-conditional-on-representation component; cells = stated category $\times$ within-game belief terciles (categories alone in the DG); bootstrap SEs in parentheses; representation shares are point estimates, dashed where the total effect is a negligible +0.008 (SE 0.020).

Results. Switching to actual-choice beliefs changes little (representation shares 16%→11% for UG Market, 59%→64% for TG Market, 52%→68% for UG Aid; DG unchanged by construction), in directions congenial to the paper’s reading—the market frame in the UG moves behavior mainly *within* representation cells, while beliefs carry more of the trust-game market effect.

7 Similarity between Participants’ Story Summaries and Categories

Post-Meeting Email from NG: Per Aid e Bonus, potremmo usare il riassunto dei partecipanti e vedere a quale categoria è più simile. [...] Un primo modo di usare questi dati è di validation: e i riassunti di Aid sono in aggregato più vicini a “Self-Interest” che quelli di Bonus, confermiamo che non solo nelle storie come sono scritte, ma anche nella mente dei partecipanti, la categoria di Self-Interest è più saliente dopo Aid che dopo Bonus. Un secondo modo,

più ambizioso, è di usare queste storie come misura di heterogeneity. Se in Aid un partecipante descrive la storia in maniera simile a “Self Interest” ed un altro a “Cooperation”, allora è più probabile che questo ultimo participant trasferisca più del primo e sulla base di una “cooperation reason”, e magari specialmente se il gioco è il TG piuttosto che il DG.

Methodology. Each summary is rated on three separate 0–100 scales for how strongly the retelling emphasizes: (F) fairness—another party’s claim, need, or contribution, how something should be divided; (O) the protagonist’s own position—own needs, hardship, scarcity, what they keep or protect; (J) joint outcomes and relationships—working together, joint success, benefits to both sides. The wording is neutral; the correspondence $F \rightarrow \text{Moral}$, $O \rightarrow \text{Self-interest}$, $J \rightarrow \text{Mutual Benefit/Cooperation}$ is applied afterwards and never shown to the rater. The rater (Opus 4.8) sees one summary per query and nothing else—no story name, game, condition, or behavioral variable—and reports which aspect dominates.

Result 1: the validation use succeeds, strongly.

Mean Emphasis (0–100)	F (Fairness)	O (Own Position)	J (Joint)
Aid Summaries ($N = 2,004$)	49.4	45.1	29.4
Bonus Summaries ($N = 2,003$)	58.1	33.9	43.1
Aid – Bonus	-8.7 ($t = -20$)	$+11.2$ ($t = 29$)	-13.8 ($t = -43$)

Aid retellings emphasize the protagonist’s own position; Bonus retellings emphasize the joint outcome and, secondarily, fairness. This is the direction of the similarity rating of the story *texts* (Aid +2.5 toward the Self-interest vignettes, Bonus +2.1 toward Mutual Benefit), but an order of magnitude larger: the difference in what the stories make salient is bigger in participants’ own retellings than in the texts as rated.

Result 2: the heterogeneity use fails. Within a story condition, summary emphasis predicts essentially nothing downstream. The dominant aspect matches the later stated reason category in 54.4% of cases against roughly 51% under independence. In the within-cell regressions of actions on the emphasis scores, 13 of 16 coefficients are insignificant at the 5% level, and the significant ones have inconsistent signs (in the Aid ultimatum cell, *own-position* emphasis enters positively)—the pattern one expects from noise. Beliefs show no relationship anywhere. Splitting senders at the median of joint-gains emphasis, the difference in trust-game sends is +1.2 points of the endowment ($p = 0.49$). One caveat: the same split shows +4.4 points in DG-KW ($p = 0.001$), but that comparison pools the two stories and Bonus retellings have high joint emphasis by construction, so it largely reflects the Aid–Bonus treatment effect rather than within-story heterogeneity; the clean within-story tests are the insignificant ones.